Two Views of a Spring

Force view

- Hooke's law: $F = -kx$.
- The force points to the origin.

Energy view

- Stored energy: $U = \frac{1}{2}kx^2$.
- The energy is zero at $x = 0$.

Measured Quantities

Quantity	Symbol	Value
Mass	m	0.50 kg
Stiffness	k	8.0 N/m
Frequency	$\omega = \sqrt{k/m}$	4.0 rad/s
Period	$T = 2\pi/\omega$	1.57 s